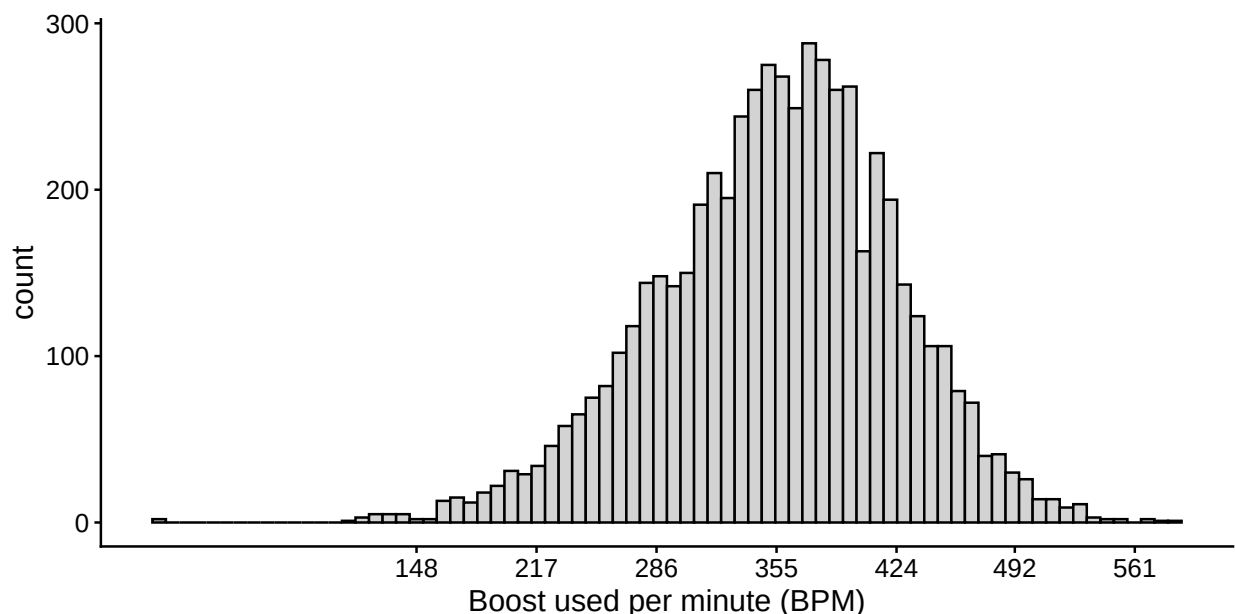


Homework 2 Solutions

(1) Rocket League is a popular online video game and E-sport that emerged in 2015. Players from around the world compete in sports like soccer (football), basketball and hockey while controlling RC-like vehicles. The most popular game-mode is doubles soccer, in which teams of two players each try to outscore each other by hitting a soccer ball into the opposing goal. Each car carries a limited supply of **boost**, which is spent to accelerate and to fly, and which refills by driving over boost pads on the field. A player's **boost used per minute (BPM)** measures how aggressively they spend boost over the course of a match. The histogram below shows the distribution of BPM for $n = 5,714$ player-match records collected from ranked matches.



- (a) Assuming a mean of $\bar{x} = 355$ and standard deviation of $s = 69$, in approximately what proportion of matches was more than 424 BPM used?

$$z = \frac{424 - 355}{69} = 1$$

Therefore we want the proportion of observations with a value greater than $\bar{x} + 1s$. Under the empirical rule 68% of observations fall within $1s$ of the mean, leaving 32% split evenly between the two tails, so the upper tail is $13.5\% + 2.5\% = 16\%$

- (b) Approximately what proportion of matches had a BPM greater than 217 but less than 493?

$217 = \bar{x} - 2s$ and $493 = \bar{x} + 2s$, so by the empirical rule approximately 95%

- (c) How much boost per minute would a player need to use in a match to be in the top 2.5% of matches?

The empirical rule puts 95% of observations within $2s$ of the mean, so 2.5% sit above $\bar{x} + 2s$. The player would need at least $355 + 2(69) = 493$ BPM

- (d) In a new match a player records a BPM of 562. Calculate the z -score for this match and explain what the z -score means in relation to the average BPM. How many standard deviations from the mean is this performance?

$$z = \frac{562 - 355}{69} = 3.00$$

This match sits exactly 3 standard deviations **above** the mean BPM. Under the empirical rule roughly 99.7% of matches fall within 3 standard deviations of the mean, so only about 0.15% of matches use this much boost or more - an extremely aggressive game.

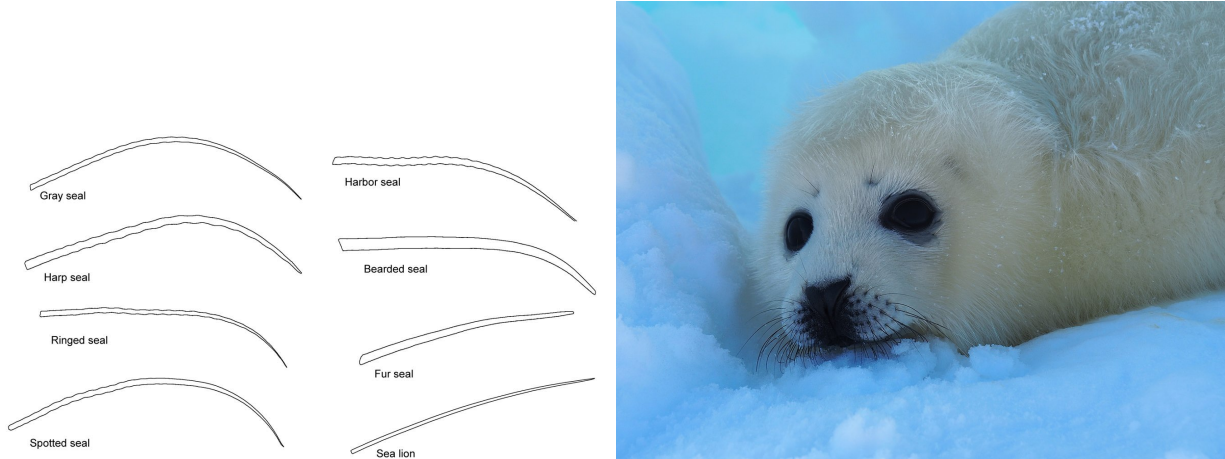


Figure 1: Pinniped whisker morphology - Ginter et al. 2012 (left), and a juvenile Harp Seal (right).

(2) Pinnipeds (seals and sea lions) possess the largest vibrissae (whiskers) among mammals and their vibrissal hair shafts demonstrate a diversity of shapes (**See Figure 1**). In a study conducted by (Ginter et al. 2012), researchers measured 9 characteristics of Pinniped whiskers in individuals from 9 species of seals and sea lions. Their goal was to better characterize whisker morphology and evolution. The following data are the recorded whisker lengths of 20 Harp Seals (given in **Table 1**). Note that the data have been sorted according to increasing “Total Length (cm)” for your convenience - use the table to answer the following questions:

Table 1: Whisker Length data for $n = 20$ Harp Seals

Observation	Whisker Total Length (cm)	Number of Beads
1	4.18	2
2	4.23	2
3	4.31	2
4	4.31	2
5	4.33	2
6	4.34	2
7	4.35	2
8	4.38	2
9	4.40	2
10	4.41	2
11	4.43	3
12	4.44	2
13	4.46	2
14	4.47	3
15	4.48	3
16	4.51	2
17	4.59	3
18	4.60	2
19	4.62	2
20	5.08	3

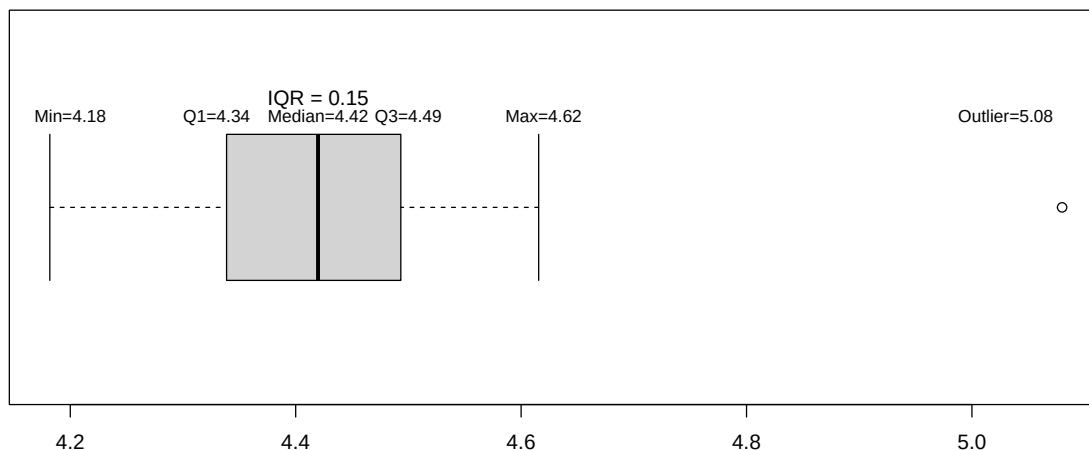
- (a) Compute the quartiles, interquartile range, mean, and standard deviation for the variable “Total Length (cm)”. Use the values exactly as they appear in **Table 1**. Remember that we find the quartiles by the **median of halves** method used in class: find the median first, and when n is odd leave that median out of both halves.

With $n = 20$ the median is the average of the 10th and 11th values, and each half has 10 values, so $Q1$ is the average of the 5th and 6th and $Q3$ the average of the 15th and 16th.

$$\begin{aligned}
 Q1 &= \frac{4.33 + 4.34}{2} = 4.335 \\
 median &= \frac{4.41 + 4.43}{2} = 4.42 \\
 Q3 &= \frac{4.48 + 4.51}{2} = 4.495 \\
 IQR &= 4.495 - 4.335 = 0.16 \\
 \bar{x} &= \frac{4.18 + 4.23 + \cdots + 5.08}{20} = 4.446 \\
 s &= \sqrt{\frac{(4.18 - 4.446)^2 + (4.23 - 4.446)^2 + \cdots + (5.08 - 4.446)^2}{19}} = 0.189
 \end{aligned} \tag{1}$$

- (b) Make a boxplot of the variable “Total Length” and make note of any outliers. Use the $1.5 \times IQR$ rule to decide what counts as an outlier.

The fences are $Q1 - 1.5 \times IQR = 4.335 - 0.24 = 4.095$ and $Q3 + 1.5 \times IQR = 4.495 + 0.24 = 4.735$. Only observation 20 (5.08 cm) falls outside, so it is the single outlier.



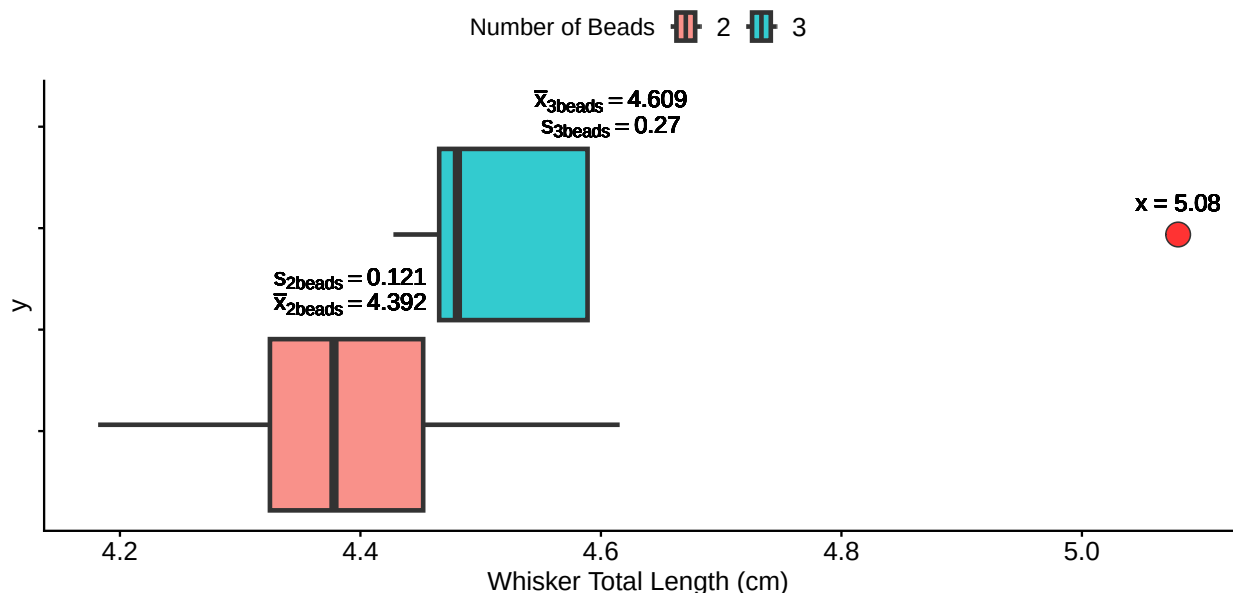
- (c) Observation 20 has a total whisker length of 5.08 (cm), how many standard deviations is this observation from the mean whisker length? (hint compute the z -score for this observation)

The sample mean is $\bar{x} = 4.446$ and the sample standard deviation is $s = 0.189$

$$z = \frac{5.08 - \bar{x}}{s} = \frac{5.080 - 4.446}{0.189} = 3.35$$

Observation 20 lies about 3.35 standard deviations above the mean whisker length.

(3) Use the following pair of boxplots constructed from the Pinniped whisker data in **Table 1**. The boxplots show the distribution of total whisker length (cm) for whiskers with 2 beads vs whiskers with three beads. Answer the following questions:



- (a) Using the boxplots above, do the whisker lengths for 2 bead and 3 bead whiskers follow a normal distribution? Why or why not? Answer separately for each group.

The boxplot for 2 bead whiskers is roughly symmetric - the median sits near the middle of the box and the two whiskers are of similar length - which is consistent with (though not proof of) an approximately normal distribution. The boxplot for 3 bead whiskers is clearly right-skewed: the group has only $n = 5$ observations and one of them (5.08 cm) sits far above the rest, dragging the upper whisker out. Those lengths are **not** approximately normal.

- (b) Suppose researchers observe two new Harp Seals (denoted Seal 1 and Seal 2) and record their total whisker lengths and the number of beads on the whiskers. Seal 1 has a total whisker length of 4.32 (cm) and the whisker has 3 beads. Seal 2 has a total whisker length of 4.34 (cm) but the whisker has only 2 beads. Suppose we want to know how they compare relative to the distribution of whiskers with the same number of beads. Confirm that the whisker length of Seal 1 has a z -score of -1.07 while Seal 2 has a z -score of -0.43 . Use the group means and standard deviations printed on the boxplot above.

Each seal must be standardized against the group with the **same** number of beads. From the figure, $\bar{x}_{2beads} = 4.392$ with $s_{2beads} = 0.121$, and $\bar{x}_{3beads} = 4.609$ with $s_{3beads} = 0.270$. Seal 1 has 3 beads, so

$$z_1 = \frac{4.32 - \bar{x}_3}{s_3} = \frac{4.32 - 4.609}{0.270} = -1.07$$

Seal 2 has 2 beads, so

$$z_2 = \frac{4.34 - \bar{x}_2}{s_2} = \frac{4.34 - 4.392}{0.121} = -0.43$$

Note that although Seal 2's whisker is the **longer** of the two in centimetres, Seal 1 sits further below its own group's mean. Comparing raw lengths across the two groups would have given the opposite impression - that is exactly what the z -score is for.

- (c) Suppose the researchers observe a new whisker with 3 beads. Using the $\pm 2s$ rule, how long or how short would that whisker have to be before you would flag it as an outlier? Then look at the 3 bead group in **Table 1** and explain why the $\pm 2s$ rule should be treated with caution here. (hint: your answer to

part (a) for the 3 bead group is the reason, and note that the $1.5 \times IQR$ rule and the $\pm 2s$ rule do not always flag the same observations)

Under the $\pm 2s$ rule an observation is flagged when $x < \bar{x} - 2s$ **or** $x > \bar{x} + 2s$, so the whisker would have to be longer than

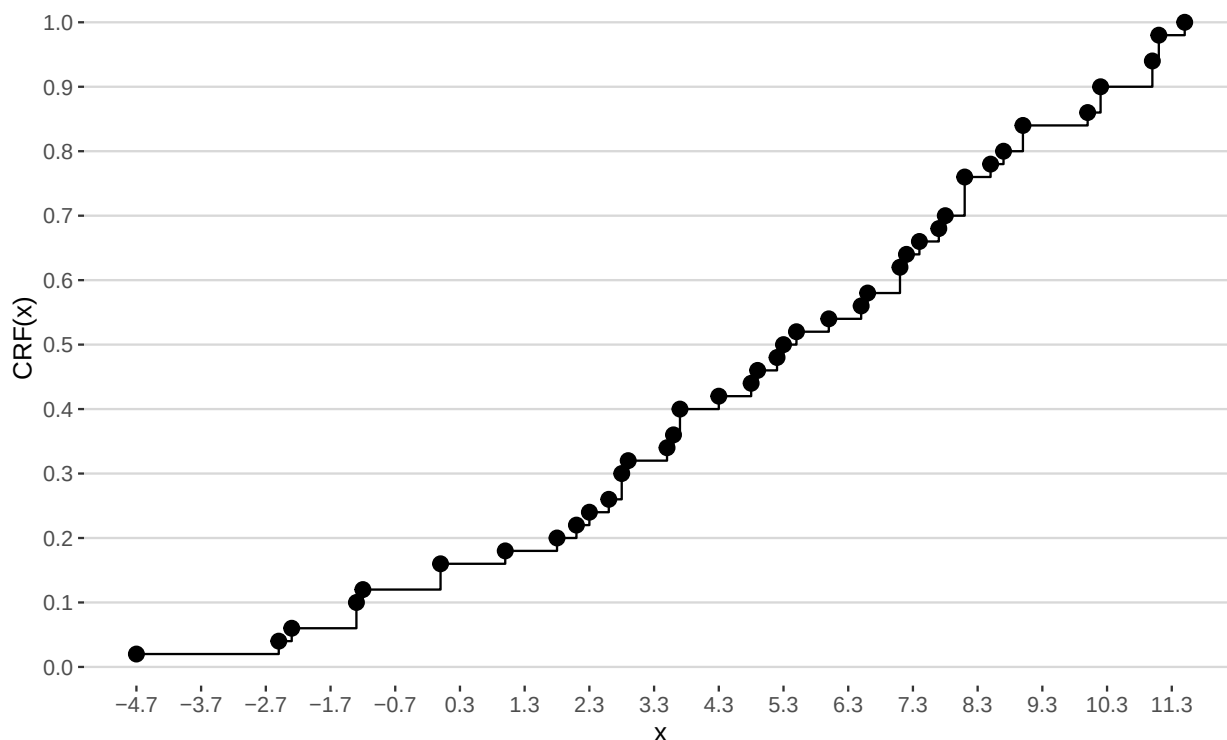
$$\bar{x}_3 + 2s_3 = 4.609 + 2 \times 0.270 = 5.149 \text{ (cm)}$$

or shorter than

$$\bar{x}_3 - 2s_3 = 4.609 - 2 \times 0.270 = 4.069 \text{ (cm)}$$

Why to be cautious: the $\pm 2s$ rule is built on the empirical rule, which assumes an approximately normal distribution - and in part (a) we decided the 3 bead group is not normal. Worse, $\bar{x}_3$ and s_3 are themselves computed from only five observations, one of which (5.08) is the extreme value we might want to detect. That single point inflates s_3 and pushes the upper cut-off out to 5.149, so the rule does **not** flag 5.08 at all - even though the $1.5 \times IQR$ rule applied to the full sample in Problem 2(b) did flag it. The two rules can disagree, and neither should be applied mechanically to a small, skewed group.

(4) Use the following plot of the cumulative distribution of a quantitative variable x to answer questions (a)-(c)



- (a) Assume the above cumulative distribution represents an approximately normal (symmetric, bell-shaped) distribution with mean $\bar{x} = 5$ and standard deviation $s = 4$. Under the empirical rule, write down the interval, in terms of $\bar{x}$ and s , that contains approximately 95% of the observations, and then give its two endpoints as numbers.

The empirical rule says approximately 95% of the observations fall in the interval $\bar{x} \pm 2s$. Substituting $\bar{x} = 5$ and $s = 4$,

$$\bar{x} - 2s = 5 - 2(4) = -3 \quad \bar{x} + 2s = 5 + 2(4) = 13$$

so the interval is $(-3, 13)$.

- (b) Your two endpoints from part (a) are the empirical rule's estimates of the 2.5th and 97.5th percentiles of X . Now read the 2.5th and 97.5th percentiles directly off the cumulative distribution plot above. Why do the two sets of answers differ? (hint: think about shape and how that relates to the empirical rule)

Reading across from $CRF(x) = 0.025$ and $CRF(x) = 0.975$ on the plot gives approximately -2.5 and 11.1 , against the -3 and 13 predicted by the empirical rule. They differ because the distribution of X is skewed to the left rather than symmetric and bell-shaped, and the empirical rule only holds for approximately normal distributions. The further a distribution departs from normality, the worse the rule's percentile estimates become.

- (c) Using the mean and standard deviation from parts a-b, compute and interpret the z -scores for the observations $x_1 = 11.5$ and $x_2 = -4.7$. Would these two observations be considered outliers under the $\pm 2s$ rule?

$$z_1 = \frac{11.5 - 5}{4} \approx 1.63$$

The observation $x_1 = 11.5$ sits 1.63 standard deviations above the mean. It is not an outlier, because $1.63 < 2$

$$z_2 = \frac{-4.7 - 5}{4} \approx -2.43$$

The observation $x_2 = -4.7$ sits 2.43 standard deviations below the mean. It would be flagged as an outlier, because $-2.43 < -2$

(5) Define the following terms

- (a) Explanatory variable - The independent variable which we manipulate to see how the response variable changes
- (b) Response variable - The outcome or dependent variable - the variable on which we make comparisons for different values of the explanatory variable
- (c) experimental study - A study which assigns subjects/observations to one or more treatments and observes the outcome of the response variable
- (d) observational study - A nonexperimental study in which the researcher observes values of the response and explanatory variables for each subject/observation
- (e) survey - A type of nonexperimental study in which subjects/observations are selected from a population for the purpose of making inferences about the population.
- (f) lurking (confounding) variable - A variable that is not among the explanatory or response variables being studied, but which is associated with both of them, and so can create the appearance of a relationship between them (or hide a real one).
- (g) In one or two sentences, explain why a statistical **association** between two variables is not by itself evidence that one **causes** the other. Give an example of a lurking variable that could produce an association without a causal link.

An association only tells us that two variables move together; it does not tell us why. In an observational study a lurking variable may be driving both, and without random assignment we cannot rule that out. For example, ice cream sales and drownings rise together, but neither causes the other - summer heat is the lurking variable that drives both. (Any sensible example is acceptable here, provided the proposed lurking variable is plausibly associated with **both** the explanatory and the response variable.)

(6) Name the sampling method used in each of the following situations:

- (a) A man is standing outside of a grocery store handing out questionnaires to shoppers asking them to evaluate their shopping experience and the quality of service. He does not ask shoppers who have their

hands full carrying groceries, but instead asks all shoppers who have only a few items or have both hands free to fill out the questionnaire.

convenience sampling

- (b) A gardener has a garden box in which he has planted 10 rows of corn. Each row consists of eight corn plants. The gardener wishes to know the average height of the corn plants in the garden box so he randomly selects rows 1, 5, and 9 and measures the height of all corn plants in those rows.

cluster sampling

- (c) A company wishes to know how employees feel about compensation and benefits. Over the next few weeks, the company asks 30 employees from each of its branch locations to fill out a survey regarding compensation.

stratified random sampling

- (d) A local polling company wishes to know what proportion of citizens in Moscow, Idaho are voting for the Republican candidate in the mayoral race. The polling company records this information for every 5th voter who leaves the voting booth.

systematic sampling

- (e) A lumber company wishes to estimate the amount of usable lumber on a plot of land. The company randomly selects 100 trees and measures their diameter, height, and volume, in order to estimate the number of board feet of each tree.

simple random sampling

- (f) Explain the difference between **sampling error** and **nonsampling error**, and give one example of each.

Sampling error is the error that arises simply because we observed a sample rather than the whole population: even a perfectly executed random sample will not match the population exactly, and a different random sample would have given a slightly different estimate. It shrinks as the sample size grows. Example: a well-run poll of 1,000 voters reports 52% support when the true population value is 54%.

Nonsampling error is every other source of error - it comes from how the study was carried out, not from the act of sampling, and it does **not** shrink as the sample size grows. Examples: undercoverage (part of the population has no chance of being selected), nonresponse, response bias from a leading question, or data entry mistakes.

- (g) A university emails a satisfaction survey to all 11,000 of its students and 600 respond. The students who reply are much more likely to be the ones with a strong complaint. Name the source of bias at work here, and explain why emailing 40,000 students instead of 11,000 would **not** fix the problem.

This is **nonresponse bias** (the sample is effectively a voluntary response sample: only those motivated to reply are represented). Increasing the number of students contacted does not help, because the same mechanism still decides who replies - the dissatisfied students remain over-represented among the respondents. A larger sample makes the estimate more **precise**, but it is precisely centred on the wrong value; a bigger sample does not remove bias. The fix is to improve the response rate or to follow up on a random subsample of the nonrespondents.

(7) Name the experimental design used in each of the following situations. Choose from **completely randomized design**, **randomized block design**, and **matched pairs design**.

- (a) A nutritionist wants to compare three breakfast cereals. She recruits 90 volunteers and uses a random number generator to assign 30 of them to each cereal.

completely randomized design

- (b) A dermatologist testing a new ointment applies the ointment to one arm of each patient and a placebo cream to the patient's other arm, deciding at random which arm gets which.

matched pairs design (each patient serves as their own control - this is the special case of a randomized block design where each block has size two)

- (c) An agronomist testing four fertilizers suspects that soil drainage strongly affects yield. He sorts his 40 plots into “well-drained” and “poorly-drained” groups, then randomly assigns the four fertilizers within each group.

randomized block design (drainage is the blocking variable)

- (d) A sleep researcher recruits 200 subjects, and a coin flip decides whether each one receives a new sleep medication or an identical-looking pill containing no medication.

completely randomized design

- (e) For the study in part (d), identify the **experimental units**, the **factor**, the **levels** of that factor, the **treatments**, and the **control group**. Was this study **blinded**? What additional information would you need to know whether it was **double blinded**?

Experimental units (subjects): the 200 recruited people. **Factor:** the sleep medication. **Levels:** two - active medication, and no medication. **Treatments:** the new sleep medication, and the placebo pill. **Control group:** the subjects who receive the identical-looking placebo pill.

The study is **blinded**, because the placebo is identical in appearance, so subjects cannot tell which treatment they received. To know whether it is **double blinded** we would also need to know whether the researchers who interact with the subjects and measure the outcome are themselves unaware of who got which treatment.

- (f) A study finds that students who attend more review sessions earn higher exam scores. The review sessions were optional. Explain what **random assignment** would have let the researchers conclude that this study cannot, and why **random sampling** alone is not enough to establish cause.

Because attendance was optional, the students who chose to attend may differ systematically from those who did not - they may be more motivated, or have more free time, and either would raise exam scores on its own. Motivation is a lurking variable, so this observational study can establish only an **association**.

Random assignment - deciding by chance which students attend - would balance those lurking variables across the two groups on average, so any difference in exam scores could be attributed to the review sessions themselves. That is what licenses a **causal** conclusion.

Random sampling does something different: it makes the sample representative, so results **generalize** to the population. It says nothing about why the two groups differ, so it cannot establish cause on its own.

(8) Define the following terms/concepts

- (a) Sample space - The set of all possible outcomes of a random trial.
- (b) Event - A subset of the sample space; a collection of one or more possible outcomes.
- (c) Random trial - A process or experiment whose individual outcome cannot be predicted in advance, but for which the set of possible outcomes is known.
- (d) Union of two events - The event $A \cup B$ that A occurs, or B occurs, or both.
- (e) Intersection of two events - The event $A \cap B$ that A **and** B both occur.
- (f) Disjoint (mutually exclusive) events - Two events that cannot both occur on the same trial, so $P(A \cap B) = 0$.
- (g) Random variable - A variable whose numerical value is determined by the outcome of a random trial.
- (h) Discrete random variable - A random variable that can take on only a countable set of values (often whole numbers), such as the number of heads in three coin flips.
- (i) Continuous random variable - A random variable that can take on any value in an interval, such as a height or a waiting time.

- (j) Probability distribution - A listing or rule that gives every possible value of a random variable together with its probability. The probabilities are each between 0 and 1 and sum to 1.
- (k) The Law of Large Numbers - As the number of trials increases, the relative frequency with which an event occurs converges to the true probability of that event.
- (9) Answer the following probability questions. Show your work.
- (a) A fair coin is thrown 3 times. What is the probability that at least one heads is observed? (hint: it is easier to find the probability of the complement first)

The complement of “at least one heads” is “no heads at all”, i.e. three tails:

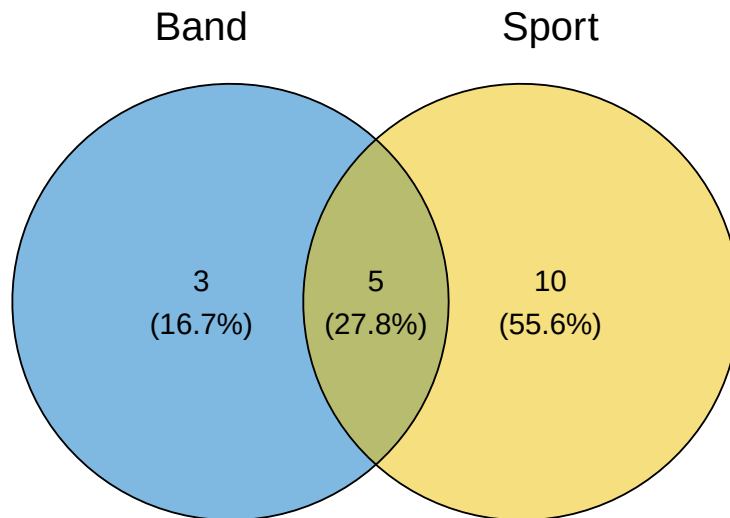
$$P(\text{no heads}) = 0.5 \times 0.5 \times 0.5 = 0.125$$

By the complement rule,

$$P(\text{at least one heads}) = 1 - 0.125 = 0.875$$

- (b) Suppose that in a class of 30 students, 8 students are in band, 15 students play a sport, and 5 students are both in band and play a sport. Let A be the event that a student is in band and let B be the event that a student plays a sport. Draw a Venn diagram that models this situation, filling in the number of students in each region. What is the probability that a student is in the band **or** plays a sport?

The Venn diagram has $8 - 5 = 3$ students in band only, 5 in the overlap, $15 - 5 = 10$ in sport only, and $30 - (3 + 5 + 10) = 12$ outside both circles.



Because the two events are **not** disjoint (5 students are in both), we must subtract the overlap so it is not counted twice - this is the addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{8}{30} + \frac{15}{30} - \frac{5}{30} = \frac{18}{30} = 0.60$$

- (c) A standard deck of 52 cards contains four suits (Hearts, Diamonds, Clubs, and Spades). Each suit contains 13 cards: 9 numbered cards, 3 face cards, and an Ace. A single card is drawn at random. What is the probability that it is a numbered card? What is the probability that it is the Ace of Spades?

There are $4 \times 9 = 36$ numbered cards, so

$$P(\text{numbered}) = \frac{36}{52} = 0.692$$

There is exactly one Ace of Spades, so

$$P(\text{Ace of Spades}) = \frac{1}{52} = 0.019$$

- (d) An urn contains 5 green balls, 7 red balls, and 4 yellow balls. Two balls are selected at random **with replacement**. What is the probability that the first ball is yellow and the second ball is red?

There are $5 + 7 + 4 = 16$ balls. With replacement the two draws are independent, so by the multiplication rule

$$P(\text{yellow, then red}) = \frac{4}{16} \times \frac{7}{16} = \frac{28}{256} = 0.109$$

- (e) Repeat part (d), but now assume the two balls are selected **without replacement**. Explain why your answer changed.

Without replacement the first ball is not returned, so only 15 balls remain for the second draw (and all 7 red balls are still among them):

$$P(\text{yellow, then red}) = \frac{4}{16} \times \frac{7}{15} = \frac{28}{240} = 0.117$$

The answer changed because the draws are no longer independent - the outcome of the first draw changes the composition of the urn, and therefore the probability of the second. Here removing a yellow ball leaves red slightly more likely, so the probability rises.

- (f) The probability that it rains R on a given day in Moscow, Idaho is $P(R) = 0.1$. The probability that it is windy W is $P(W) = 0.2$, and the probability that it is cloudy C is $P(C) = 0.7$. Assume these three weather events are independent. What is the probability that a given day is rainy, cloudy, **and** windy?

For independent events the multiplication rule extends to all three:

$$P(R \cap C \cap W) = 0.1 \times 0.7 \times 0.2 = 0.014$$

(10) Consider a survey of tree volume. Suppose that in the population of trees 10% have a volume of 10 cubic feet, 20% have a volume of 20 cubic feet, 30% have a volume of 30 cubic feet, and 40% have a volume of 40 cubic feet. Let X denote the volume of a randomly selected tree, assuming every tree has an equal chance of being selected. The probability distribution of X is given in **Table 2** below.

Table 2: Probability distribution of tree volume X

Tree Volume (cubic feet)	$P(X)$
10	0.1
20	0.2
30	0.3
40	0.4

- (a) Verify that the four probabilities in **Table 2** form a valid probability distribution.

Each probability lies between 0 and 1, and they sum to one: $0.1 + 0.2 + 0.3 + 0.4 = 1.0$. Both conditions hold, so this is a valid probability distribution.

- (b) Confirm that X has a mean of $\mu = 30$ cubic feet by computing $\mu = \sum_x x \cdot P(x)$ yourself.

$$\mu = 10(0.1) + 20(0.2) + 30(0.3) + 40(0.4) = 1 + 4 + 9 + 16 = 30 \text{ cubic feet}$$

- (c) Confirm that X has a standard deviation of $\sigma = 10$ cubic feet by computing $\sigma^2 = \sum_x (x - \mu)^2 \cdot P(x)$ and then taking the square root.

$$\begin{aligned}
\sigma^2 &= (10 - 30)^2(0.1) + (20 - 30)^2(0.2) + (30 - 30)^2(0.3) + (40 - 30)^2(0.4) \\
&= 400(0.1) + 100(0.2) + 0(0.3) + 100(0.4) \\
&= 40 + 20 + 0 + 40 = 100
\end{aligned} \tag{2}$$

$$\sigma = \sqrt{100} = 10 \text{ cubic feet}$$

(d) Is X as described here a discrete or a continuous random variable? Explain.

As described here X is **discrete**: it takes only the four listed values, and each has its own probability. (Tree volume measured in the real world would be continuous - any value in an interval is possible - but this population has been simplified to four volume categories.)

(11) Answer the following, showing the formula you used and the numbers you substituted into it.

- (a) An angler has a 40% chance of catching a fish on every cast, and each cast is independent of the others. Using the binomial probability mass function

$$P(X = k) = \frac{n!}{k!(n-k)!} \cdot p^k \cdot (1-p)^{n-k}$$

show how to compute the probability that she catches exactly 2 fish in her next 5 casts.

Here $n = 5$, $k = 2$ and $p = 0.4$:

$$P(X = 2) = \frac{5!}{2!(5-2)!} (0.4)^2 (0.6)^3 = 10 \times 0.16 \times 0.216 = 0.3456$$

So there is about a 34.6% chance she catches exactly two fish.

- (b) For the angler in part (a), what is the mean number of fish she expects to catch in 5 casts, and what is the variance? (hint: $E[X] = np$ and $Var[X] = np(1-p)$)

$$E[X] = np = 5(0.4) = 2 \text{ fish} \quad Var[X] = np(1-p) = 5(0.4)(0.6) = 1.2$$

- (c) A small campus coffee stand serves on average 3 customers per 10-minute period. Let X be the number of customers in a 10-minute period, and assume X follows a Poisson distribution with $\lambda = 3$. Using

$$P(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

compute the probability that **no** customers arrive in a given 10-minute period.

With $\lambda = 3$ and $k = 0$, and recalling that $0! = 1$ and $3^0 = 1$:

$$P(X = 0) = \frac{3^0 e^{-3}}{0!} = e^{-3} = 0.0498$$

So about a 5% chance of no customers.

- (d) For the coffee stand in part (c), compute the probability that **at most one** customer arrives in a given 10-minute period.

“At most one” means $k = 0$ or $k = 1$, so add the two probabilities:

$$P(X = 1) = \frac{3^1 e^{-3}}{1!} = 3e^{-3} = 0.1494$$

$$P(X \leq 1) = P(X = 0) + P(X = 1) = 0.0498 + 0.1494 = 0.1992$$

So about a 20% chance of one customer or fewer.

References

References

Ginter, Carly C, Thomas J DeWitt, Frank E Fish, and Christopher D Marshall. 2012. “Fused Traditional and Geometric Morphometrics Demonstrate Pinniped Whisker Diversity.” *PloS One* 7 (4): e34481.